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BSIT DA 2-1
CC 104

A. Discuss Briefly

1. What is an algorithm, and how is it related to the tasks you perform in your daily life?

An algorithm is defined as a series of instructions or procedures used in solving a problem or undertaking an activity. Algorithms are used in our everyday activities, especially when we engage in doing things that involve a procedure. For example, we may cook food, get ready for school, and use the telephone.

2. Why is it important for an IT student to learn algorithms?

Learning algorithms is essential for an IT student as algorithms improve the thinking ability and problem solving skill. Algorithms help students gain knowledge about how computer programs function and how to create an efficient program.

B. Convert each JavaScript code into a step-by-step algorithm using simple English or pseudocode.

1. "for (let i = 1; i < 10; i *= 2)"

1. Start with "i = 1".
2. Check if "i" is less than 10.
3. If true, display the value of "i".
4. Multiply "i" by 2.
5. Repeat the process while "i" is less than 10.
6. Stop when "i" reaches or becomes greater than 10.

Output: "1, 2, 4, 8"

2. "for (let i = 0; i < 10; i++)"

1. Start with "i = 0".
2. Check if "i" is less than 10.
3. If true, display the value of "i".
4. Add 1 to "i".
5. Repeat the process while "i" is less than 10.
6. Stop when "i" reaches 10.

Output: "0, 1, 2, 3, 4, 5, 6, 7, 8, 9"

C. Run both JavaScript codes. Based on the outputs and execution of the two codes, answer the following:

1. What did you observe about the sequence of numbers produced by the first code?

The first code produces numbers that double each time: 1, 2, 4, 8. The numbers increase quickly because the value is multiplied by 2 after every loop.

2. What did you observe about the sequence of numbers produced by the second code?

The second code produces consecutive numbers starting from 0 up to 9. The value increases by 1 after every loop.

3. What is the main difference between the way the two loops progress?

The first loop increases by multiplying the value by 2, while the second loop increases by adding 1 each time.

4. How does the way an algorithm performs its steps affect its output?

The way an algorithm performs its steps directly affects its output. If the steps use different operations or are performed in a different order, the program can produce different results. In these examples, multiplying by 2 creates a doubling sequence, while adding 1 creates a consecutive sequence.